

Homework 3

ECE 253 - Fall 2025

Problem 1

Consider the following 4-bit image (intensity range: 0–15):

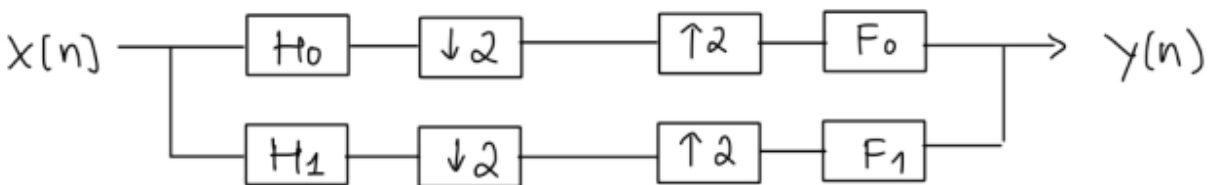
$$\begin{bmatrix} 2 & 2 & 2 & 5 & 5 & 9 \\ 2 & 2 & 2 & 5 & 5 & 9 \\ 8 & 8 & 8 & 5 & 5 & 9 \\ 7 & 7 & 7 & 7 & 7 & 9 \\ 7 & 7 & 7 & 7 & 7 & 9 \\ 8 & 8 & 8 & 9 & 9 & 9 \end{bmatrix}$$

- (a) Perform **global histogram equalization** on the image.
- (b) Perform **local histogram equalization** using a local window of size 3×3 .
- (c) Compare and discuss the results.

In parts (a) and (b), show your work regarding the computation of the histogram, cumulative distribution function (CDF), intensity mappings, and the final histogram of the results.

Problem 2

Consider a two-channel filter bank:



- (a) Let H_0 and F_0 be ideal low-pass filters with a cutoff frequency at $\pi/2$, and H_1 and F_1 be ideal high-pass filters with a cutoff frequency at $\pi/2$. Verify that the overall system is a perfect reconstruction (PR) system.

Hint: Use frequency domain analysis for a particular choice of $X(e^{j\omega})$.

(b) Let

$$H_0(z) = \frac{1}{\sqrt{2}}(1 + z^{-1}), \quad H_1(z) = \frac{1}{\sqrt{2}}(1 - z^{-1}).$$

Find $F_0(z)$ and $F_1(z)$ such that aliasing is canceled and verify that this is a PR system.

Problem 3

Consider an FIR perfect reconstruction (PR) filter bank satisfying the condition:

$$F_0(z)H_0(z) - F_0(-z)H_0(-z) = 2z^{-L}.$$

- (a) Let $F_0(z) = H_0(z)$. What is the necessary PR condition in terms of $H_0(z)$?
- (b) Find all possible solutions for $H_0(z)$.
- (c) What are the corresponding filters $H_1(z)$, $F_0(z)$ and $F_1(z)$?